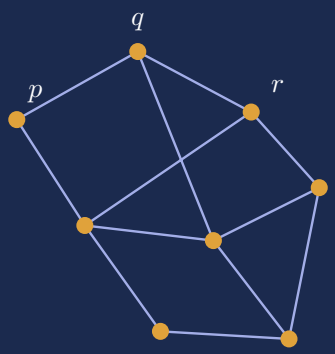
$\therefore \quad \Rightarrow \quad \Leftrightarrow \quad \neg \quad \forall \quad \exists \quad \blacksquare$



DISCRETE MATHEMATICS AND GRAPH THEORY

Unit 4
Proof Techniques

Some terminology · Proof methods and strategies · Forward proof · Proof by contradiction · Proof by contraposition · Proof of necessity and sufficiency

LECTURE NOTES · SOLVED PROBLEMS · EXERCISES WITH SOLUTIONS

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Syllabus hours	4 hrs (Unit 4.0)
Prepared for	B.Tech CSE students, GEC Munger
Exam pattern	University theory paper: 2, 5/7 and 10/15-mark questions

References: K. H. Rosen, *Discrete Mathematics and its Applications*; S. S. Epp, *Discrete Mathematics with Applications*; C. L. Liu & D. P. Mohapatra, *Elements of Discrete Mathematics*; J. P. Tremblay & R. Manohar, *Discrete Mathematical Structures with Applications to Computer Science*; N. L. Biggs, *Discrete Mathematics*; S. Lipschutz & M. Lipson, *Discrete Mathematics* (Schaum's Outlines).

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How to use these notes

Every topic follows the same pattern: **definition** (teal) → **key facts** (indigo) → **worked examples** → **pitfalls** (red) and **exam tips** (green). Sections 1 and 3–7 end with an exercise set whose complete solutions follow at once in teal boxes. Section 8 shows common mistakes in proofs; Section 9 has solved university examination questions (the source is given with each); Section 10 is a graded exercise set (Parts A–C) whose complete solutions are in Section 11. A one-page revision sheet closes the notes.

Notation: $\neg p$ = not p; $p \to q$ = if p then q; $\Rightarrow$ = implies; $\iff$ = if and only if; $a \mid b$ = a divides b; $\mathbb{Z}$ = integers, $\mathbb{Q}$ = rational numbers, $\mathbb{R}$ = real numbers; ■ = end of proof.

Syllabus item (Unit 4.0)	Section	Suggested hours
Some Terminology	1	0.5
Proof Methods and Strategies	2, 7, 8	1.0
Forward Proof	3	0.5
Proof by Contradiction	4	0.75
Proof by Contraposition	5	0.5
Proof of Necessity and Sufficiency	6	0.75

Unit 3 (Propositional Logic) gives the logic behind every method used here: the conditional $p \to q$, its contrapositive, the rules of inference and the quantifiers.

1 Introduction and Some Terminology

A **proof** is a step-by-step argument that shows a mathematical statement is true. Each step must follow from the hypotheses, from definitions, from basic facts (axioms) or from results already proved, using the rules of inference of Unit 3. Proofs matter in computer science too: we prove that an algorithm always gives the correct answer, that a program always stops, or that a security protocol cannot be broken in a certain way.

1.1 Basic terms

Term	Meaning
Statement (proposition)	A sentence that is either true or false, but not both.
Definition	An exact description of the meaning of a word, e.g. "an integer n is *even* if $n = 2k$ for some integer k".
Axiom (postulate)	A statement accepted as true without proof; it is the starting point of a theory.
Theorem	An important statement that has been *proved* to be true.
Proposition	A less important theorem (the word is also used for "statement").
Lemma	A small theorem proved mainly to help in proving a bigger theorem.
Corollary	A result that follows easily (in a few lines) from a theorem just proved.
Conjecture	A statement believed to be true but *not yet proved*. Once proved, it becomes a theorem; if disproved, it is false.
Proof	A valid argument that shows a theorem is true.
Hypothesis, conclusion	In a theorem of the form "if p then q", p is the **hypothesis** (premise) and q is the **conclusion**.
Counterexample	A single example that shows a "for all" statement is false.
Q.E.D. or ■	"*Quod erat demonstrandum*" — "which was to be proved"; marks the end of a proof.

Example 1.1 – Examples of each term

Give an example of an axiom, a theorem, a lemma, a corollary and a conjecture.

Solution.

- **Axioms:** for all real numbers a and b, $a + b = b + a$; through two distinct points there passes exactly one straight line.
- **Theorem:** there are infinitely many prime numbers (proved in Section 4). Pythagoras' theorem is another example.
- **Lemma:** "if n^2 is even, then n is even". On its own it is not very exciting, but it is the key step in proving that $\sqrt{2}$ is irrational.
- **Corollary:** the theorem "the angles of a triangle add up to $180°$" has the corollary "each angle of an equilateral triangle is $60°$".
- **Conjectures:** *Goldbach's conjecture* — every even integer greater than 2 is the sum of two primes ($4 = 2+2, 6 = 3+3, 8 = 3+5, \ldots$). It has been checked by computer for very large numbers but has never been proved. *Fermat's Last Theorem* ($x^n + y^n = z^n$ has no solution in positive integers for $n > 2$) was a conjecture for about 350 years until Andrew Wiles proved it in 1994; now it is a theorem.

1.2 Definitions used in this unit

Most proofs in this unit use the following definitions. A proof must go back to the definition; "it is obvious" is not a reason.

Definition: Even, odd, divides, prime, rational

Let a, b, n be integers.

- n is **even** if $n = 2k$ for some integer k; n is **odd** if $n = 2k + 1$ for some integer k. Every integer is either even or odd, but not both.
- a **divides** b (written $a \mid b$) if $b = ak$ for some integer k.
- An integer $p > 1$ is **prime** if its only positive divisors are 1 and p; an integer $n > 1$ that is not prime is **composite**.
- n is a **perfect square** if $n = k^2$ for some integer k.
- A real number r is **rational** if $r = \dfrac{a}{b}$ for some integers a, b with $b \neq 0$; otherwise r is **irrational**. Every rational number can be written in *lowest terms*, i.e. with a and b having no common factor other than 1.

Parity rule	Reason	Result
even + even	$2k + 2l = 2(k + l)$	even
odd + odd	$(2k + 1) + (2l + 1) = 2(k + l + 1)$	even
even + odd	$2k + (2l + 1) = 2(k + l) + 1$	odd
even × any	$2k \cdot m = 2(km)$	even
odd × odd	$(2k + 1)(2l + 1) = 2(2kl + k + l) + 1$	odd

1.3 How theorems are stated

Many theorems hide a "for all". For example, "If $x > y$, where x and y are positive real numbers, then $x^2 > y^2$" really means

$$\forall x\, \forall y\, [(x > 0) \wedge (y > 0) \wedge (x > y) \to (x^2 > y^2)].$$

To prove such a statement we take **arbitrary** (any, not special) elements x and y of the universe, prove the conditional for them, and then use universal generalization (Unit 3, Section 7) to conclude that it holds for all x and y. Choosing particular numbers, such as $x = 3$, $y = 2$, shows only one case and is *not* a proof.

Example 1.2 – Hypothesis and conclusion

For each theorem, write the hypothesis and the conclusion, and the statement in symbols.
(i) The square of an odd integer is odd. (ii) If $a \mid b$ and $b \mid c$, then $a \mid c$. (iii) An integer is even only if its square is even.

Solution. (i) Hypothesis: n is an odd integer; conclusion: n^2 is odd. In symbols: $\forall n \in \mathbb{Z}\, [\, \text{odd}(n) \to \text{odd}(n^2)\,]$.
(ii) Hypothesis: $a \mid b$ and $b \mid c$; conclusion: $a \mid c$. In symbols: $\forall a, b, c\, [\, (a \mid b) \wedge (b \mid c) \to (a \mid c)\,]$.
(iii) "p only if q" means $p \to q$. Hypothesis: n is even; conclusion: n^2 is even. In symbols: $\forall n \in \mathbb{Z}\, [\, \text{even}(n) \to \text{even}(n^2)\,]$.

Example 1.3 – Theorem, lemma or conjecture?

Classify each statement: (i) "Every even integer greater than 2 is the sum of two primes." (ii) "If n^2 is even, then n is even", used as a step in proving that $\sqrt{2}$ is irrational. (iii) "$\sqrt{2}$ is irrational." (iv) "The sum of two rational numbers is rational, so the sum of any finite number of rational numbers is rational."

Solution. (i) A conjecture (Goldbach's conjecture): not yet proved. (ii) A lemma: a small result used to prove a bigger one. (iii) A theorem. (iv) The first part is a theorem; "so the sum of any finite number of rational numbers is rational" is a corollary that follows from it easily.

Examination tip

A 2-mark question such as "Define lemma and corollary" is answered best with the definition *and* one example each.

1.4 Exercise Set 1A — terminology

1A.1 Define: (i) axiom (ii) theorem (iii) lemma (iv) corollary (v) conjecture. Give one example of each.

1A.2 What is the difference between a theorem and a conjecture? Can a conjecture become a theorem?

1A.3 Write the hypothesis and the conclusion of: (i) "If two angles of a triangle are equal, then the triangle is isosceles." (ii) "A number is divisible by 6 only if it is divisible by 3." (iii) "$x^2 = 4$ whenever $x = -2$."

1A.4 Rewrite with the hidden quantifiers shown: (i) "The product of two odd integers is odd." (ii) "If x is a real number, then $x^2 \geq 0$."

1A.5 Using the definitions, decide which are true and give a one-line reason: (i) 0 is even. (ii) -7 is odd. (iii) $3 \mid 0$. (iv) 0.75 is rational. (v) 1 is prime.

1.5 Solutions to Exercise Set 1A

1A.1 – Definitions with examples

(i) **Axiom:** a statement accepted as true without proof, e.g. $a+b = b+a$ for real numbers. (ii) **Theorem:** an important statement that has been proved, e.g. there are infinitely many primes. (iii) **Lemma:** a small result used to prove a bigger one, e.g. "if n^2 is even then n is even", used to prove $\sqrt{2}$ irrational. (iv) **Corollary:** a result that follows easily from a theorem, e.g. each angle of an equilateral triangle is 60° (from "the angles of a triangle add up to 180°"). (v) **Conjecture:** a statement believed true but not yet proved, e.g. Goldbach's conjecture.

1A.2 – Theorem versus conjecture

A theorem has a proof; a conjecture does not (yet). A conjecture becomes a theorem when someone proves it — Fermat's Last Theorem was a conjecture until it was proved in 1994. A conjecture can also be shown false by a counterexample.

1A.3 – Hypothesis and conclusion

(i) Hypothesis: two angles of a triangle are equal; conclusion: the triangle is isosceles.
(ii) "p only if q" is $p \to q$. Hypothesis: the number is divisible by 6; conclusion: it is divisible by 3.
(iii) "q whenever p" is $p \to q$. Hypothesis: $x = -2$; conclusion: $x^2 = 4$.

1A.4 – Hidden quantifiers

(i) For all integers m and n, if m and n are odd, then mn is odd: $\forall m\, \forall n\, [\, \text{odd}(m) \wedge \text{odd}(n) \to \text{odd}(mn)\,]$.
(ii) $\forall x \in \mathbb{R}\, [\, x^2 \geq 0\,]$.

1A.5 – Using the definitions

(i) True: $0 = 2 \cdot 0$. (ii) True: $-7 = 2(-4) + 1$. (iii) True: $0 = 3 \cdot 0$. (iv) True: $0.75 = \frac{3}{4}$. (v) False: a prime must be greater than 1.

2 Proof Methods and Strategies

Most theorems have the form $p \to q$ (often with a hidden "for all"). A conditional $p \to q$ is false only when p is true and q is false. Every method of proof shows that this case *cannot* happen; a **disproof** shows that it *does* happen.

2.1 The methods at a glance

Method	What you do	Why it works
Forward (direct) proof	assume p; deduce q	$p \to q$ can be false only when p is true
Proof by contradiction	assume p and $\neg q$; reach a contradiction	rule of contradiction: $(\neg u \to F_0) \Rightarrow u$
Proof by contraposition	assume $\neg q$; deduce $\neg p$	$p \to q \equiv \neg q \to \neg p$
Necessity and sufficiency	prove $p \to q$ and $q \to p$	$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
Proof by cases	split p into $p_1 \vee p_2$; prove each case	$(p_1 \vee p_2) \to q \equiv (p_1 \to q) \wedge (p_2 \to q)$
Proof by exhaustion	check every element of a small finite set	$\forall$ on a finite set is a finite "and"
Proof of existence	show one a with $p(a)$	one example makes $\exists x\, p(x)$ true
Disproof by counterexample	show one a with $p(a)$ false	one example makes $\forall x\, p(x)$ false

2.2 Choosing a method

If the statement looks like ...	try first ...
"if p then q" and p gives an equation to work with (e.g. $n = 2k + 1$)	forward (direct) proof
"if p then q" but $\neg q$ is easier to use than p (e.g. "if n^2 is odd then n is odd")	proof by contraposition
a negative statement: "no ...", "...is irrational", "there are infinitely many ..."	proof by contradiction
"p if and only if q", "necessary and sufficient", "the following are equivalent"	prove both directions (Section 6)
the hypothesis splits naturally (even/odd, positive/negative)	proof by cases
"there exists ..."	give an example (proof of existence)
"for all ..." and you suspect it is false	look for a counterexample

2.3 Strategy: forward and backward reasoning

- **Forward reasoning:** start from the hypothesis and ask "what follows from this?" Use definitions and known facts to move step by step towards the conclusion.
- **Backward reasoning:** start from the conclusion and ask "what would be enough to prove this?" Keep going back until you reach something you already know is true.

In practice we often find the steps by working backwards, and then *write* the proof forwards (Section 3).

2.4 Vacuous and trivial proofs

Definition: Vacuous and trivial proofs

Vacuous proof: if the hypothesis p is false, then $p \to q$ is true whatever q is.
Trivial proof: if the conclusion q is true, then $p \to q$ is true whatever p is.

Example 2.1 – A vacuous proof

Let $P(n)$ be "if $n > 1$, then $n^2 > n$". Show that $P(0)$ is true.

Solution. $P(0)$ is "if $0 > 1$, then $0^2 > 0$". The hypothesis $0 > 1$ is false, so the conditional is true. This is a vacuous proof.

Example 2.2 – A trivial proof

Let $P(n)$ be "if a and b are positive integers with $a \geq b$, then $a^n \geq b^n$". Show that $P(0)$ is true.

Solution. $P(0)$ says "if $a \geq b$, then $a^0 \geq b^0$". Since $a^0 = b^0 = 1$, the conclusion $1 \geq 1$ is true, so the conditional is true without using the hypothesis. This is a trivial proof.

2.5 How to write a proof

A simple template

1. **State the method** ("We give a proof by contradiction.").
2. **Write what you assume** (the hypothesis, or the negation for contraposition/contradiction), using the definitions: "Let n be odd, so $n = 2k + 1$ for some integer k."
3. **Give the steps**, each with its reason (algebra, a definition, an earlier result).
4. **State what has been shown** and end with ■.

Common pitfall

Do not start from the conclusion and work towards something true, then stop: that shows only that the conclusion *implies* a true statement, which proves nothing. If you find the steps backwards, rewrite them in the forward direction (and check that every step can be reversed).

3 Forward Proof (Direct Proof)

Definition: Forward (direct) proof of $p \to q$

1. Hypothesis: assume that p is true. **2. Analysis:** starting from p, and using definitions, known facts and the rules of logic, move forward step by step until you reach q. **3. Conclusion:** $p \to q$ is true.

Why this is enough: $p \to q$ can be false only when p is true and q is false. A forward proof shows that whenever p is true, q is true as well, so that case never happens.

Example 3.1 – The square of an odd integer

Give a direct proof of the statement "The square of an odd integer is an odd integer."

Solution. The statement is "for all $n \in \mathbb{Z}$, if n is odd then n^2 is odd". Let n be any odd integer; then $n = 2k + 1$ for some integer k. Hence

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which has the form $2m + 1$ with $m = 2k^2 + 2k$ an integer. So n^2 is odd. Since n was arbitrary, the statement holds for all odd integers. ■

Example 3.2 – Sum and product of two odd integers

Give a direct proof of: "For all integers k and l, if k and l are both odd, then $k + l$ is even and kl is odd."

Solution. Take any two integers k and l, and assume both are odd. Then $k = 2m + 1$, $l = 2n + 1$ for some integers m, n. Therefore

$$k + l = 2(m + n + 1) \quad \text{(even)}, \qquad kl = 4mn + 2(m + n) + 1 = 2(2mn + m + n) + 1 \quad \text{(odd)}.$$

Since k and l were arbitrary, the statement holds for all integers. ■

Example 3.3 – Product of two perfect squares

Prove that if m and n are both perfect squares, then mn is also a perfect square.

Solution. Assume m and n are perfect squares. Then $m = s^2$ and $n = t^2$ for some integers s, t. Hence $mn = s^2t^2 = (st)^2$, and st is an integer. So mn is a perfect square. ■

Example 3.4 – Product of two rational numbers

Prove that the product of two rational numbers is rational.

Solution. Let r and s be rational. Then $r = \frac{a}{b}$ and $s = \frac{c}{d}$ with integers a, b, c, d and $b \neq 0, d \neq 0$. So

$$rs = \frac{ac}{bd},$$

where ac and bd are integers and $bd \neq 0$ (a product of two non-zero integers is non-zero). Hence rs is rational. ■

Example 3.5 – Divisibility is transitive

Let a, b, c be integers. Prove that if $a \mid b$ and $b \mid c$, then $a \mid c$.

Solution. Assume $a \mid b$ and $b \mid c$. By definition, $b = ak$ and $c = bl$ for some integers k, l. Then $c = (ak)l = a(kl)$, and kl is an integer, so $a \mid c$. ■

Example 3.6 – Backward reasoning, forward proof

Prove that for any two distinct positive real numbers x and y,

$$\frac{x+y}{2} > \sqrt{xy} \qquad \text{(arithmetic mean} > \text{geometric mean)}.$$

Solution. *Finding the proof (backwards).* Both sides are positive, so the inequality holds if its square holds:

$$\frac{(x+y)^2}{4} > xy \iff (x+y)^2 > 4xy \iff x^2+2xy+y^2-4xy > 0 \iff (x-y)^2 > 0.$$

The last statement is true because $x \neq y$.
Writing the proof (forwards). Since $x \neq y$, $(x-y)^2 > 0$, i.e. $x^2 - 2xy + y^2 > 0$. Adding $4xy$ to both sides gives $(x+y)^2 > 4xy$, so $\frac{(x+y)^2}{4} > xy$. Taking positive square roots, $\frac{x+y}{2} > \sqrt{xy}$. ■

Example 3.7 – Squares of odd integers and 8

Prove that if n is an odd integer, then $n^2 - 1$ is divisible by 8.

Solution. Let $n = 2k+1$. Then $n^2 - 1 = 4k^2 + 4k = 4k(k+1)$. Of the two consecutive integers k and $k+1$, one is even, so $k(k+1) = 2j$ for some integer j. Hence $n^2 - 1 = 4 \cdot 2j = 8j$, which is divisible by 8. ■ (Check: $3^2 - 1 = 8$, $5^2 - 1 = 24$, $7^2 - 1 = 48$.)

Example 3.8 – A quadratic that is always positive

Prove that $x^2 - 4x + 5 > 0$ for every real number x.

Solution. For any real x, $x^2 - 4x + 5 = (x^2 - 4x + 4) + 1 = (x-2)^2 + 1$. Since $(x-2)^2 \geq 0$, we get $(x-2)^2 + 1 \geq 1 > 0$. ■

Common pitfall

Checking examples ($3^2 = 9$ is odd, $5^2 = 25$ is odd, ...) is *not* a proof of a "for all" statement. A proof must work for an *arbitrary* element, written in general form such as $n = 2k+1$.

3.1 Exercise Set 3A — forward proofs

Give a direct (forward) proof of each statement.

3A.1 If m is an even integer, then $m + 7$ is an odd integer.

3A.2 The square of an even integer is an even integer.

3A.3 The sum of two odd integers is an even integer.

3A.4 For all integers k and l, if k and l are both even, then (a) $k + l$ is even and (b) kl is even.

3A.5 If an integer a is such that $a - 2$ is divisible by 3, then $a^2 - 1$ is divisible by 3.

3A.6 The sum of any five consecutive integers is divisible by 5.

3A.7 For all real numbers x and y, $x^2 + y^2 \geq 2xy$.

3A.8 If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

3.2 Solutions to Exercise Set 3A

3A.1–3A.4 – Parity

3A.1: $m = 2k \Rightarrow m + 7 = 2k + 7 = 2(k + 3) + 1$, which is odd.
3A.2: $n = 2k \Rightarrow n^2 = 4k^2 = 2(2k^2)$, which is even.
3A.3: $a = 2k + 1, b = 2l + 1 \Rightarrow a + b = 2(k + l + 1)$, which is even.
3A.4: $k = 2m, l = 2n \Rightarrow$ (a) $k + l = 2(m + n)$, even; (b) $kl = 4mn = 2(2mn)$, even.
In each case the integers were arbitrary, so the statement holds for all of them. ■

3A.5–3A.6 – Divisibility

3A.5: $a - 2 = 3k \Rightarrow a = 3k + 2 \Rightarrow a^2 - 1 = (a - 1)(a + 1) = (3k + 1)(3k + 3) = 3(3k + 1)(k + 1)$, which is divisible by 3.
3A.6: five consecutive integers are $n, n + 1, n + 2, n + 3, n + 4$; their sum is $5n + 10 = 5(n + 2)$, which is divisible by 5. ■

3A.7–3A.8

3A.7: For any real x, y, $(x-y)^2 \geq 0$, i.e. $x^2-2xy+y^2 \geq 0$. Adding $2xy$ to both sides gives $x^2+y^2 \geq 2xy$.
3A.8: $b = ak$ and $c = al$ for integers k, l. Then $b + c = a(k + l)$, so $a \mid (b + c)$. ■

4 Proof by Contradiction

Definition: Proof by contradiction (*reductio ad absurdum*)

To prove a statement P, assume that P is *false* and show that this leads to a contradiction — something that is impossible, such as $r \wedge \neg r$, or a fact we know is false (e.g. $1 = 2$). Since the assumption leads to something impossible, the assumption is wrong, so P is true.

For a conditional $p \to q$: **1. Hypothesis:** assume p is true *and* q is false. **2. Analysis:** using definitions, facts and the rules of logic, reach a contradiction. **3. Conclusion:** $p \to q$ is true.

Why it works. This is the rule of contradiction (Unit 3, Section 6): $(\neg P \to F_0) \Rightarrow P$. If $\neg P$ leads to something always false, then $\neg P$ must be false.

Example 4.1 – If n^2 is odd then n is odd

Prove by contradiction: "For every integer n, if n^2 is odd, then n is odd."

Solution. Suppose the statement is false: for some integer n, n^2 is odd and n is not odd, i.e. n is even, say $n = 2k$. Then $n^2 = 4k^2 = 2(2k^2)$ is even. So n^2 is both odd and even — a contradiction. Hence the statement is true. ■

Example 4.2 – The square of an even integer

Prove by contradiction that the square of an even integer is an even integer.

Solution. Assume the opposite: n is even and n^2 is odd. Then $n^2 = n \cdot n$ is not divisible by 2, so n is not divisible by 2, i.e. n is not even — contradicting the assumption that n is even. Hence the statement is true. ■

Example 4.3 – If m is even then $m+7$ is odd

Prove by contradiction: if m is an even integer, then $m+7$ is an odd integer.

Solution. Assume m is even and $m+7$ is even, say $m+7=2k$. Then $m=2k-7=2(k-4)+1$, so m is odd — a contradiction, since m is even. Hence the statement is true. ■

Example 4.4 – $x+y\geq 100$

Prove by contradiction: "For all real numbers x and y, if $x+y\geq 100$, then $x\geq 50$ or $y\geq 50$."

Solution. Assume the statement is false for some x, y: $x+y\geq 100$, but $x<50$ and $y<50$. Adding the last two inequalities gives $x+y<100$, which contradicts $x+y\geq 100$. Hence the statement is true for all x, y. ■

Example 4.5 – $\sqrt{2}$ is irrational

Prove that there is no rational number whose square is 2, i.e. that $\sqrt{2}$ is irrational.

Solution. Assume the opposite: $\sqrt{2}$ is rational. Then $\sqrt{2}=\frac{a}{b}$ where a, b are integers, $b\neq 0$, and the fraction is in *lowest terms* (a and b have no common factor other than 1).

Step 1. Squaring, $2=\frac{a^2}{b^2}$, so $a^2=2b^2$. Hence a^2 is even.

Step 2. Then a is even: if a were odd, a^2 would be odd (Example 3.1). So $a=2c$ for some integer c.

Step 3. Substituting, $4c^2=2b^2$, so $b^2=2c^2$. Hence b^2 is even, and (as in Step 2) b is even.

Step 4. Now a and b are both even, so 2 is a common factor — this contradicts "lowest terms".

Conclusion. The assumption is false, so $\sqrt{2}$ is irrational. ■

Example 4.6 – There are infinitely many primes

Prove that there are infinitely many prime numbers.

Solution. Assume the opposite: there are only finitely many primes, say $p_1, p_2, \ldots, p_n$. Consider

$$Q=p_1p_2\cdots p_n+1.$$

$Q>1$, so Q has at least one prime factor p. This p must be one of $p_1, \ldots, p_n$, so p divides the product $p_1p_2\cdots p_n$. Since p also divides Q, it divides the difference $Q-p_1p_2\cdots p_n=1$. But no prime divides 1 — a contradiction. Hence there are infinitely many primes. ■

Example 4.7 – Twenty-two days

Show that at least four of any 22 days must fall on the same day of the week.

Solution. Assume the opposite: at most three of the 22 days fall on each day of the week. There are 7 days in a week, so at most $7\times 3=21$ days could be chosen — a contradiction, since we have 22 days. Hence at least four of them fall on the same day of the week. ■

Example 4.8 – $a^2-4b\neq 2$

Prove that for all integers a and b, $a^2-4b\neq 2$.

Solution. Assume the opposite: $a^2-4b=2$ for some integers a, b. Then $a^2=4b+2=2(2b+1)$ is even, so a is even (if a were odd, a^2 would be odd, by Example 3.1): $a=2c$. Then $4c^2-4b=2$, i.e. $2c^2-2b=1$. The left side is even and the right side is odd — a contradiction. Hence $a^2-4b\neq 2$. ■

4.1 Disproof by contradiction

Definition: Disproof by contradiction

To show that $p \to q$ is *false*, assume that p and q are both true and show that this leads to a contradiction. Then q cannot be true when p is true, so $p \to q$ is false.

Example 4.9 – The sum of two odd integers

Disprove the statement "The sum of two odd integers is an odd integer."

Solution. Assume a and b are odd and $a+b$ is odd: $a = 2k_1+1$, $b = 2k_2+1$ and $a+b = 2k_3+1$ for some integers k_1, k_2, k_3. But $a+b = 2(k_1+k_2+1)$ is even. So $a+b$ is both odd and even — a contradiction. Hence the statement is false. (For example, $3+5=8$.)

Examination tip

In a proof by contradiction, always begin with the words "Assume, to the contrary, that ..." and write the negation correctly. The negation of "$\sqrt{2}$ is irrational" is "$\sqrt{2}$ is rational"; the negation of "there are infinitely many primes" is "there are only finitely many primes".

4.2 Exercise Set 4A — proof by contradiction

4A.1 Prove by contradiction: (i) If m is an even integer, then $m+7$ is an odd integer. (ii) For any real number x, if $x^2 > 0$, then $x \neq 0$. (iii) If x and y are integers such that xy is odd, then x and y are both odd. (iv) If x is a rational number and y is an irrational number, then $x+y$ is an irrational number. (v) If n is the product of two positive integers a and b, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

4A.2 Disprove by contradiction: (i) The square of an odd integer is an even integer. (ii) If the heights of two triangles are equal, then their areas are equal. (iii) The roots of a cubic equation with real coefficients can all be complex (non-real).

4A.3 Prove that $\sqrt{5}$ is irrational.

4A.4 Prove that $\log_2 3$ is irrational.

4A.5 If 25 pens are put into 6 boxes, show that some box contains at least 5 pens.

4.3 Solutions to Exercise Set 4A

4A.1 – Five proofs by contradiction

(i) Assume m is even and $m+7$ is not odd, $m+7 = 2k$. Then $m = 2(k-4)+1$ is odd — contradiction.
(ii) Assume $x^2 > 0$ and $x = 0$. Then $x^2 = 0$, contradicting $x^2 > 0$.
(iii) Assume xy is odd and x or y is even. If $x = 2k$ then $xy = 2(ky)$ is even; similarly if y is even — contradicting "xy is odd".
(iv) Assume x is rational, y is irrational and $x+y = r$ is rational. Then $y = r-x$ is a difference of two rationals, hence rational — contradicting "y is irrational".
(v) Assume $n = ab$ with $a > \sqrt{n}$ and $b > \sqrt{n}$. Then $n = ab > \sqrt{n}\cdot\sqrt{n} = n$, i.e. $n > n$ — a contradiction.
Each assumption "p true and q false" leads to a contradiction, so each $p \to q$ is true. ■

4A.2 – Disproofs by contradiction

(i) Assume n is odd and n^2 is even: $n = 2k+1$ and $n^2 = 2m$. But $n^2 = 4k^2+4k+1 = 2(2k^2+2k)+1$ is odd, so n^2 is both even and odd — a contradiction. Hence the statement is false.

(ii) Assume the statement is true. Then any two triangles with the same height h and bases b_1, b_2 satisfy $\frac{1}{2}b_1h = \frac{1}{2}b_2h$, so $b_1 = b_2$: triangles of equal height would always have equal bases. But the triangles with $h = 1$, $b_1 = 2$ and $h = 1$, $b_2 = 4$ have equal heights and unequal bases (areas 1 and 2) — a contradiction. Hence the statement is false.
(iii) Assume a cubic $ax^3 + bx^2 + cx + d = 0$ with real coefficients ($a \neq 0$) has all three roots non-real. Non-real roots of a polynomial with real coefficients occur in conjugate pairs $\alpha, \bar{\alpha}$, so the number of non-real roots (counted with multiplicity) is even. But three is odd — a contradiction. Hence at least one root is real, and the statement is false. ■

4A.3 – $\sqrt{5}$ is irrational

Assume $\sqrt{5} = \frac{a}{b}$ in lowest terms. Then $a^2 = 5b^2$, so $5 \mid a^2$. Since 5 is prime, $5 \mid a$ (if a prime divides a product, it divides one of the factors): $a = 5c$. Then $25c^2 = 5b^2$, so $b^2 = 5c^2$ and, in the same way, $5 \mid b$. Now 5 is a common factor of a and b — a contradiction. Hence $\sqrt{5}$ is irrational. ■

4A.4 – $\log_2 3$ is irrational

Assume $\log_2 3 = \frac{a}{b}$ with integers a, b, $b > 0$. Since $\log_2 3 > 0$, also $a > 0$. Then $2^{a/b} = 3$, so $2^a = 3^b$. The left side is even (as $a \geq 1$) and the right side is odd — a contradiction. Hence $\log_2 3$ is irrational. ■

4A.5 – 25 pens in 6 boxes

Assume every box contains at most 4 pens. Then the 6 boxes hold at most $6 \times 4 = 24$ pens — a contradiction, since there are 25 pens. Hence some box contains at least 5 pens. ■

5 Proof by Contraposition

Definition: Proof by contraposition (indirect proof)

The **contrapositive** of $p \to q$ is $\neg q \to \neg p$. Since $p \to q \equiv \neg q \to \neg p$, we may prove $p \to q$ by giving a forward proof of $\neg q \to \neg p$:
1. Assume q is false. **2.** Deduce, step by step, that p is false. **3.** Conclude that $\neg q \to \neg p$, and therefore $p \to q$, is true.

The two statements always have the same truth value:

p	q	$p \to q$	$\neg q$	$\neg p$	$\neg q \to \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

When to use it. Use contraposition when the conclusion q is hard to start from but its negation $\neg q$ gives an equation to work with. For example, in "if n^2 is odd then n is odd", the hypothesis $n^2 = 2k + 1$ tells us little about n, but $\neg q$ ("n is even", $n = 2k$) is easy to use.

Example 5.1 – If n^2 is odd then n is odd

Let n be an integer. Prove that if n^2 is odd, then n is odd.

Solution. We prove the contrapositive: *if n is not odd, then n^2 is not odd.* Suppose n is even: $n = 2k$ for some integer k. Then $n^2 = 4k^2 = 2(2k^2)$, so n^2 is even, i.e. not odd. The contrapositive is true, so the

statement is true. ■

Example 5.2 – If n^2 is even then n is even

Let n be an integer. Prove that if n^2 is even, then n is even.

Solution. Contrapositive: *if n is odd, then n^2 is odd.* This is exactly Example 3.1: $n = 2k+1 \Rightarrow n^2 = 2(2k^2+2k)+1$. Hence the statement is true. ■ (This is the lemma used in the proof that $\sqrt{2}$ is irrational.)

Example 5.3 – Product of two even integers

Give an indirect proof of "If a and b are even integers, then ab is even."

Solution. Contrapositive: *if ab is not even, then a and b are not both even.* Suppose ab is odd. If a were even, $a = 2k$, then $ab = 2(kb)$ would be even; so a is not even. In the same way b is not even. Hence a and b are not both even. The contrapositive is true, so the statement is true. ■

Example 5.4 – If $xy > 25$ then $x > 5$ or $y > 5$

Give an indirect proof of: "For all positive real numbers x and y, if the product xy exceeds 25, then $x > 5$ or $y > 5$."

Solution. Let p: $xy > 25$, q: $x > 5$, r: $y > 5$. The statement is $p \to (q \vee r)$; its contrapositive is $\neg(q \vee r) \to \neg p \equiv (\neg q \wedge \neg r) \to \neg p$. Take any positive real numbers x, y with $x \le 5$ and $y \le 5$. Then $xy \le 5 \times 5 = 25$, i.e. xy does not exceed 25. So the contrapositive is true for all positive reals x, y, and therefore so is the statement. ■

Example 5.5 – Stating the contrapositive first

Give an indirect proof of each statement, stating its contrapositive first:
(i) For all integers k and l, if kl is odd, then both k and l are odd.
(ii) For all integers k and l, if $k + l$ is even, then k and l are both even or both odd.

Solution. (i) *Contrapositive:* for all integers k, l, if k is even or l is even, then kl is even. If $k = 2m$, then $kl = 2(ml)$ is even; similarly if $l = 2n$, $kl = 2(kn)$ is even. The contrapositive is true, so the statement is true.
(ii) *Contrapositive:* for all integers k, l, if one of k, l is odd and the other is even, then $k + l$ is odd. Say $k = 2m + 1, l = 2n$; then $k + l = 2(m + n) + 1$ is odd (similarly with the roles exchanged). Hence the statement is true. ■

Example 5.6 – If ab is even then a or b is even

Prove that for integers a and b, if ab is even, then a is even or b is even.

Solution. Contrapositive: *if a and b are both odd, then ab is odd.* Let $a = 2k + 1$, $b = 2l + 1$. Then $ab = 4kl + 2k + 2l + 1 = 2(2kl + k + l) + 1$ is odd. Hence the statement is true. ■

Example 5.7 – One statement, three proofs

Prove "If n is an odd integer, then $n + 9$ is an even integer" by (i) forward proof (ii) contraposition (iii) contradiction.

Solution. Let p: n is odd, q: $n + 9$ is even.
(i) *Forward:* assume n is odd, $n = 2k + 1$. Then $n + 9 = 2k + 10 = 2(k + 5)$ is even.
(ii) *Contraposition:* assume $n + 9$ is not even, i.e. $n + 9 = 2k + 1$. Then $n = 2k - 8 = 2(k - 4)$ is even, i.e. not odd. So $\neg q \to \neg p$, hence $p \to q$.
(iii) *Contradiction:* assume n is odd and $n + 9$ is odd, $n + 9 = 2k + 1$. Then $n = 2(k - 4)$ is even, contradicting "n is odd". Hence $p \to q$. ■

5.1 Contraposition, contradiction and the converse

To prove $p \to q$ by	You assume	You must reach
Forward proof	p	q
Contraposition	$\neg q$	$\neg p$
Contradiction	p and $\neg q$	any contradiction (something impossible)

Every proof by contraposition can be turned into a proof by contradiction: assume p and $\neg q$; the contraposition argument gives $\neg p$, which contradicts p.

Common pitfall

The **converse** $q \to p$ is *not* equivalent to $p \to q$; proving the converse proves nothing about the original statement. Also take care with compound parts: the contrapositive of $p \to (q \vee r)$ is $(\neg q \wedge \neg r) \to \neg p$, *not* $(\neg q \vee \neg r) \to \neg p$.

5.2 Exercise Set 5A — proof by contraposition

Give a proof by contraposition of each statement.

5A.1 If m is an even integer, then $m + 7$ is an odd integer.

5A.2 For any real number x, if $x^2 > 0$, then $x \neq 0$.

5A.3 If x and y are integers such that xy is odd, then x and y are both odd.

5A.4 If x is a rational number and y is an irrational number, then $x + y$ is an irrational number.

5A.5 If n is the product of two positive integers a and b, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

5A.6 If n is an integer and $5n + 4$ is odd, then n is odd.

5A.7 If x and y are real numbers with $x + y \geq 2$, then $x \geq 1$ or $y \geq 1$.

5.3 Solutions to Exercise Set 5A

5A.1–5A.3

5A.1: *Contrapositive:* if $m + 7$ is not odd, then m is not even. If $m + 7 = 2k$, then $m = 2k - 7 = 2(k - 4) + 1$ is odd, i.e. not even.
5A.2: *Contrapositive:* if $x = 0$, then x^2 is not greater than 0. Indeed $x = 0 \Rightarrow x^2 = 0$.
5A.3: *Contrapositive:* if x is even or y is even, then xy is even. If $x = 2k$, $xy = 2(ky)$; if $y = 2l$, $xy = 2(xl)$. Either way xy is even, i.e. not odd. ■

5A.4–5A.5

5A.4: Let p: x is rational and y is irrational; q: $x + y$ is irrational. *Contrapositive* $\neg q \to \neg p$: if $x + y$ is rational, then x is irrational or y is rational. Suppose $x + y = r$ is rational. If x is irrational we are done; if x is rational, then $y = r - x$ is a difference of two rationals, hence rational. So $\neg p$ holds.
5A.5: *Contrapositive:* if $a > \sqrt{n}$ and $b > \sqrt{n}$, then $n \neq ab$. Indeed $ab > \sqrt{n} \cdot \sqrt{n} = n$, so $ab \neq n$. ■

5A.6–5A.7

5A.6: *Contrapositive:* if n is even, then $5n + 4$ is even. If $n = 2k$, then $5n + 4 = 10k + 4 = 2(5k + 2)$, which is even.
5A.7: *Contrapositive:* if $x < 1$ and $y < 1$, then $x + y < 2$. Adding the two inequalities gives $x + y < 1 + 1 = 2$. ■

6 Proof of Necessity and Sufficiency

6.1 Necessary and sufficient conditions

Recall from Unit 3 that the following all mean $p \to q$:

"if p then q", "p is **sufficient** for q", "q is **necessary** for p", "p only if q".

For example, "being a square is sufficient for a quadrilateral to be a rectangle" and "being a rectangle is necessary for a quadrilateral to be a square" both mean: *if a quadrilateral is a square, then it is a rectangle.*

A statement "p **if and only if** q" (written $p \leftrightarrow q$, or "p iff q", or "p is necessary and sufficient for q") says that $p \to q$ *and* $q \to p$ are both true.

Proving $p \leftrightarrow q$

Since $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$, a proof has **two parts**, and each part may use any method (forward, contraposition or contradiction):

Statement	Part	What to prove
"p if and only if q"	"if" part	$q \to p$
	"only if" part	$p \to q$
"p is necessary and sufficient for q"	sufficiency	$p \to q$ (p is enough for q)
	necessity	$q \to p$ (q cannot happen without p)

Example 6.1 – n is even iff n^2 is even

Prove that an integer n is even if and only if n^2 is even.

Solution. *($\Rightarrow$, forward)* If n is even, $n = 2k$, then $n^2 = 2(2k^2)$ is even.
($\Leftarrow$, contraposition) If n is odd, $n = 2k+1$, then $n^2 = 2(2k^2+2k)+1$ is odd (Example 3.1). So "n^2 even $\Rightarrow$ n even".
Both directions hold, so n is even $\iff$ n^2 is even. ■

Example 6.2 – n is even iff $7n+4$ is even

Prove that for an integer n, n is even if and only if $7n+4$ is even.

Solution. *($\Rightarrow$)* If $n = 2k$, then $7n+4 = 14k+4 = 2(7k+2)$, which is even.
($\Leftarrow$, contraposition) If n is odd, $n = 2k+1$, then $7n+4 = 14k+11 = 2(7k+5)+1$, which is odd.
So "$7n+4$ even $\Rightarrow$ n even".
Hence n is even iff $7n+4$ is even. ■

Example 6.3 – Multiples of 3

Prove that for an integer n, n^2 is a multiple of 3 if and only if n is a multiple of 3.

Solution. *(Sufficiency: $3 \mid n \Rightarrow 3 \mid n^2$)* If $n = 3k$, then $n^2 = 9k^2 = 3(3k^2)$.
(Necessity: $3 \mid n^2 \Rightarrow 3 \mid n$, by contraposition) Suppose $3 \nmid n$. Then $n = 3k+1$ or $n = 3k+2$.

- If $n = 3k+1$: $n^2 = 9k^2+6k+1 = 3(3k^2+2k)+1$.
- If $n = 3k+2$: $n^2 = 9k^2+12k+4 = 3(3k^2+4k+1)+1$.

In both cases n^2 leaves remainder 1 on division by 3, so $3 \nmid n^2$. Hence $3 \mid n^2 \iff 3 \mid n$. ■

Example 6.4 – A chain of reversible steps

Prove that for a real number x: $x^2 - 3x + 2 = 0$ if and only if $x = 1$ or $x = 2$.

Solution. Each step below can be read in both directions:

$$x^2 - 3x + 2 = 0 \iff (x-1)(x-2) = 0 \iff x - 1 = 0 \text{ or } x - 2 = 0 \iff x = 1 \text{ or } x = 2.$$

(The middle step uses: a product of real numbers is zero exactly when one of the factors is zero.) ■

Common pitfall

A chain of " $\iff$ " is a valid proof only if *every* step can be reversed. Squaring cannot: $x = 3 \Rightarrow x^2 = 9$ is true, but $x^2 = 9 \Rightarrow x = 3$ is false ($x = -3$). When a step goes one way only, prove the two directions separately.

Example 6.5 – Equality in the AM–GM inequality

Prove that for positive real numbers x and y, $\dfrac{x+y}{2} = \sqrt{xy}$ if and only if $x = y$.

Solution. *($\Leftarrow$)* If $x = y$, then $\frac{x+y}{2} = x$ and $\sqrt{xy} = \sqrt{x^2} = x$ (as $x > 0$). So the two sides are equal.
($\Rightarrow$) If $\frac{x+y}{2} = \sqrt{xy}$, squaring gives $(x+y)^2 = 4xy$, so $x^2 - 2xy + y^2 = 0$, i.e. $(x-y)^2 = 0$. Hence $x = y$. ■

Example 6.6 – Necessary but not sufficient

For an integer n, is "n is divisible by 2" a necessary condition, a sufficient condition, or both, for "n is divisible by 4"?

Solution. *Necessary:* if $4 \mid n$, then $n = 4k = 2(2k)$, so $2 \mid n$. *Not sufficient:* $n = 2$ is divisible by 2 but not by 4 (a counterexample). So divisibility by 2 is necessary but not sufficient for divisibility by 4.

6.2 Several equivalent statements

To show that $p_1, p_2, \ldots, p_n$ are all equivalent ("the following are equivalent"), it is enough to prove a **cycle**

$$p_1 \to p_2, \quad p_2 \to p_3, \quad \ldots, \quad p_{n-1} \to p_n, \quad p_n \to p_1.$$

Then any p_i leads round the cycle to any other p_j. For three statements this needs only three proofs instead of six.

Example 6.7 – Three equivalent statements

Show that the following statements about an integer n are equivalent: p_1: n is even; p_2: $n - 1$ is odd; p_3: n^2 is even.

Solution. We prove $p_1 \to p_2$, $p_2 \to p_3$ and $p_3 \to p_1$.
$p_1 \to p_2$. If $n = 2k$, then $n - 1 = 2k - 1 = 2(k-1) + 1$ is odd.
$p_2 \to p_3$. If $n - 1 = 2k + 1$, then $n = 2k + 2 = 2(k+1)$, so $n^2 = 4(k+1)^2 = 2 \cdot 2(k+1)^2$ is even.
$p_3 \to p_1$. This is Example 5.2 (contraposition).

Conclusion. The three statements are equivalent. ■

Common pitfall

Proving only one direction of an "if and only if" earns at most half marks. Write the two parts under clear headings, e.g. "Sufficiency" and "Necessity", or "($\Rightarrow$)" and "($\Leftarrow$)".

6.3 Exercise Set 6A — necessity and sufficiency

6A.1 Prove that an integer n is odd if and only if $5n + 6$ is odd.

6A.2 Prove that for integers a and b, $a + b$ is even if and only if a and b are both even or both odd.

6A.3 Show that the following statements about an integer n are equivalent: (i) n is odd; (ii) $1 - n$ is even; (iii) n^2 is odd.

6A.4 For a real number x, is "$x > 2$" a necessary condition, a sufficient condition, or both, for "$x^2 > 4$"? Justify.

6A.5 Prove that for a real number x, $x^2 - 5x + 6 < 0$ if and only if $2 < x < 3$.

6.4 Solutions to Exercise Set 6A

6A.1 – n odd iff $5n + 6$ odd

($\Rightarrow$) If $n = 2k + 1$, then $5n + 6 = 10k + 11 = 2(5k + 5) + 1$, which is odd.
($\Leftarrow$, contraposition) If $n = 2k$, then $5n + 6 = 10k + 6 = 2(5k + 3)$, which is even.
Hence n is odd iff $5n + 6$ is odd. ■

6A.2 – $a + b$ even

($\Leftarrow$) If both are even, $a+b = 2k+2l = 2(k+l)$; if both are odd, $a+b = (2k+1)+(2l+1) = 2(k+l+1)$. Either way $a + b$ is even.
($\Rightarrow$, contraposition) If one is even and the other odd, say $a = 2k$, $b = 2l + 1$, then $a + b = 2(k + l) + 1$ is odd.
Hence $a + b$ is even iff a, b are both even or both odd. ■

6A.3 – Three equivalent statements

(i)$\to$(ii): $n = 2k + 1 \Rightarrow 1 - n = -2k = 2(-k)$, even.
(ii)$\to$(iii): $1 - n = 2k \Rightarrow n = 1 - 2k \Rightarrow n^2 = 1 - 4k + 4k^2 = 2(2k^2 - 2k) + 1$, odd.
(iii)$\to$(i), contraposition: if n is even, $n = 2k$, then $n^2 = 2(2k^2)$ is even, i.e. not odd.
The cycle is complete, so the three statements are equivalent. ■

6A.4 – $x > 2$ and $x^2 > 4$

Sufficient: if $x > 2$, then $x^2 = x \cdot x > 2 \cdot 2 = 4$ (both factors are positive and greater than 2).
Not necessary: $x = -3$ gives $x^2 = 9 > 4$ although $x \not> 2$.
So "$x > 2$" is sufficient but not necessary for "$x^2 > 4$".

6A.5 – $x^2 - 5x + 6 < 0$

$x^2 - 5x + 6 = (x - 2)(x - 3)$. A product of two real numbers is negative exactly when one factor is positive and the other negative.

- $x - 2 > 0$ and $x - 3 < 0 \iff 2 < x < 3$.
- $x - 2 < 0$ and $x - 3 > 0 \iff x < 2$ and $x > 3$, which is impossible.

Hence $(x - 2)(x - 3) < 0 \iff 2 < x < 3$. ■

7 More Proof Methods: Cases, Exhaustion, Existence, Uniqueness, Counterexamples

7.1 Proof by cases

Definition: Proof by cases

If the hypothesis can be split into cases $p_1 \vee p_2 \vee \ldots \vee p_n$ that together cover every possibility, prove the conclusion separately in each case. This works because

$$(p_1 \vee p_2 \vee \ldots \vee p_n) \to q \;\equiv\; (p_1 \to q) \wedge (p_2 \to q) \wedge \ldots \wedge (p_n \to q).$$

Example 7.1 – $n^2 \geq n$

Prove that $n^2 \geq n$ for every integer n.

Solution. *Case 1:* $n = 0$. Then $n^2 = 0 = n$.
Case 2: $n \geq 1$. Multiplying $n \geq 1$ by the positive number n gives $n^2 \geq n$.
Case 3: $n \leq -1$. Then $n < 0$, while $n^2 \geq 0$, so $n^2 > n$.
The three cases cover all integers, so $n^2 \geq n$ always. ■

Example 7.2 – $n^2 + n$ is even

Prove that $n^2 + n$ is even for every integer n.

Solution. *Case 1:* n even, $n = 2k$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, even.
Case 2: n odd, $n = 2k + 1$. Then $n^2 + n = (4k^2 + 4k + 1) + (2k + 1) = 2(2k^2 + 3k + 1)$, even.
Hence $n^2 + n$ is even for every integer n. ■

Example 7.3 – Squares have the form $4k$ or $4k + 1$

Prove that the square of any integer has the form $4k$ or $4k + 1$.

Solution. *Case 1:* $n = 2m$. Then $n^2 = 4m^2$, of the form $4k$.
Case 2: $n = 2m + 1$. Then $n^2 = 4m^2 + 4m + 1 = 4(m^2 + m) + 1$, of the form $4k + 1$. ■
(So no perfect square has the form $4k + 2$ or $4k + 3$; for example 6, 7, 10, 11 are not squares.)

Example 7.4 – Without loss of generality

Prove that if x and y are integers and both xy and $x + y$ are even, then x and y are both even.

Solution. We prove the contrapositive: if x or y is odd, then xy is odd or $x + y$ is odd. Because the statement is symmetric in x and y, we may assume *without loss of generality* that x is odd, $x = 2m + 1$.
Case 1: y even, $y = 2n$. Then $x + y = 2(m + n) + 1$ is odd.
Case 2: y odd, $y = 2n + 1$. Then $xy = 2(2mn + m + n) + 1$ is odd.
In both cases the contrapositive holds, so the statement is true. ■

7.2 Proof by exhaustion

Definition: Proof by exhaustion

A statement $\forall x \in S,\, p(x)$ with S *finite* can be proved by checking $p(a)$ for each $a \in S$ in turn. This is practical only when S is small. (It is a proof by cases with one case for each element.)

Example 7.5 – $(n+1)^3 \geq 3^n$ for small n

Prove that $(n+1)^3 \geq 3^n$ if n is a positive integer with $n \leq 4$.

Solution. Check each case: $n=1$: $8 \geq 3$; $n=2$: $27 \geq 9$; $n=3$: $64 \geq 27$; $n=4$: $125 \geq 81$. All four cases hold. ■ (For $n=5$ it fails: $216 < 243$, so the restriction $n \leq 4$ is needed.)

Example 7.6 – Sums of at most three squares, $2 \leq n \leq 26$

Prove that every even integer n with $2 \leq n \leq 26$ can be written as a sum of at most three perfect squares.

Solution. Verify each case:

$$\begin{array}{llll} 2 = 1^2 + 1^2 & 4 = 2^2 & 6 = 2^2 + 1^2 + 1^2 & 8 = 2^2 + 2^2 \\ 10 = 3^2 + 1^2 & 12 = 2^2 + 2^2 + 2^2 & 14 = 3^2 + 2^2 + 1^2 & 16 = 4^2 \\ 18 = 4^2 + 1^2 + 1^2 & 20 = 4^2 + 2^2 & 22 = 3^2 + 3^2 + 2^2 & 24 = 4^2 + 2^2 + 2^2 \\ 26 = 5^2 + 1^2 & & & \end{array}$$

The statement holds for every element of the finite set, so it is proved by exhaustion. ■

7.3 Proof of existence

Definition: Existence proofs

To prove $\exists x\, p(x)$:
Constructive proof: find an actual element a with $p(a)$ true.
Non-constructive proof: show that such an element must exist without naming it (often by cases or by contradiction).

Example 7.7 – A real root

Prove that there exists a real number x such that $x^3 + 2x^2 - 5x - 6 = 0$.

Solution. For $x=-1$: $(-1)^3 + 2(-1)^2 - 5(-1) - 6 = -1 + 2 + 5 - 6 = 0$. So $x=-1$ is such a number. ■

Example 7.8 – Squares whose sum is a square

Prove that there exist positive integers m and n such that m, n and $m+n$ are all perfect squares.

Solution. Take $m = 9 = 3^2$ and $n = 16 = 4^2$; then $m+n = 25 = 5^2$. ■

Example 7.9 – The taxicab number

Show that there is a positive integer that can be written as the sum of two cubes of positive integers in two different ways.

Solution. $1729 = 10^3 + 9^3 = 1000 + 729$ and $1729 = 12^3 + 1^3 = 1728 + 1$. ■ (Constructive.)

Example 7.10 – A non-constructive proof

Show that there exist irrational numbers x and y such that x^y is rational.

Solution. Consider $\sqrt{2}^{\sqrt{2}}$. There are two cases.

Case 1: $\sqrt{2}^{\sqrt{2}}$ is rational. Then $x = y = \sqrt{2}$ work.

Case 2: $\sqrt{2}^{\sqrt{2}}$ is irrational. Take $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$. Then $x^y = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{2} = 2$, which is rational.

In either case such x, y exist, although the proof does not tell us which case is true. ■

7.4 Proof of uniqueness

Definition: Uniqueness proofs

To prove "there is exactly one x with $p(x)$":
Existence: find an x with $p(x)$. **Uniqueness:** show that if $p(x)$ and $p(y)$ are both true, then $x = y$.

Example 7.11 – A linear equation

Prove that if a and b are real numbers and $a \neq 0$, then there is a unique real number r such that $ar + b = 0$.

Solution. *Existence:* $r = -\frac{b}{a}$ works, since $a\left(-\frac{b}{a}\right) + b = -b + b = 0$.
Uniqueness: suppose $ar + b = 0$ and $as + b = 0$. Then $ar + b = as + b$, so $a(r - s) = 0$. Since $a \neq 0$, $r - s = 0$, i.e. $r = s$. ■

7.5 Disproof by counterexample

Definition: Counterexample

A statement $\forall x \in S,\ p(x)$ is false as soon as we show *one* element $a \in S$ with $p(a)$ false. Such an a is a **counterexample**.

Example 7.12 – Product of two odd integers

Disprove: "The product of any two odd integers is a perfect square."

Solution. $m = 3$ and $n = 5$ are odd but $mn = 15$ is not a perfect square.

Example 7.13 – Sum of two perfect squares

Prove or disprove: "If m and n are positive integers which are perfect squares, then $m + n$ is a perfect square."

Solution. Take $m = 9 = 3^2$ and $n = 4 = 2^2$. Then $m + n = 13$, which is not a perfect square. The statement is false.

Example 7.14 – Four squares

Prove or disprove: "The sum of the squares of any four non-zero integers is an even integer."

Solution. Take $a = b = c = 1$ and $d = 2$. Then $a^2 + b^2 + c^2 + d^2 = 1 + 1 + 1 + 4 = 7$, which is not even. The statement is false.

Example 7.15 – Sums of two squares

Is every positive integer the sum of the squares of two integers?

Solution. No. The only squares not larger than 3 are 0 and 1, and $0 + 0 = 0$, $0 + 1 = 1$, $1 + 1 = 2$; none of these is 3. So 3 is a counterexample.

Examination tip

A $\forall$-statement is disproved by *one* counterexample, but it is never proved by examples (unless the set is finite and you check every element). An $\exists$-statement is proved by *one* example.

7.6 Exercise Set 7A — cases, exhaustion, existence, counterexamples

7A.1 Prove by exhaustion: (i) Every even integer x such that $30 \le x \le 58$ can be written as a sum of at most three perfect squares. (ii) Every even integer x such that $4 \le x \le 38$ can be written as a sum of two primes.

7A.2 Disprove by counterexample: (i) For all integers n, $n^2 \neq n$. (ii) For all integers m, n, $m^2 = n^2$ if and only if $m = n$. (iii) For any real number x, $x^3 > x^2$. (iv) There is no function which is equal to its derivative. (v) If two lines in space do not intersect, then they must be parallel.

7A.3 Prove by cases that $n^3 - n$ is divisible by 3 for every integer n.

7A.4 Prove that for all real numbers x and y, $\max(x, y) + \min(x, y) = x + y$.

7A.5 Prove that there is a positive integer that equals the sum of the positive integers not exceeding it.

7.7 Solutions to Exercise Set 7A

7A.1 – Proofs by exhaustion

(i) Every even x with $30 \le x \le 58$:

$$\begin{array}{llll} 30 = 5^2+2^2+1^2 & 32 = 4^2+4^2 & 34 = 5^2+3^2 & 36 = 6^2 \\ 38 = 6^2+1^2+1^2 & 40 = 6^2+2^2 & 42 = 5^2+4^2+1^2 & 44 = 6^2+2^2+2^2 \\ 46 = 6^2+3^2+1^2 & 48 = 4^2+4^2+4^2 & 50 = 7^2+1^2 & 52 = 6^2+4^2 \\ 54 = 7^2+2^2+1^2 & 56 = 6^2+4^2+2^2 & 58 = 7^2+3^2 & \end{array}$$

(ii) Every even x with $4 \le x \le 38$:

$$\begin{array}{llllll} 4 = 2+2 & 6 = 3+3 & 8 = 3+5 & 10 = 3+7 & 12 = 5+7 & 14 = 3+11 \\ 16 = 3+13 & 18 = 5+13 & 20 = 3+17 & 22 = 3+19 & 24 = 5+19 & 26 = 3+23 \\ 28 = 5+23 & 30 = 7+23 & 32 = 3+29 & 34 = 3+31 & 36 = 5+31 & 38 = 7+31 \end{array}$$

Every element of each finite set has been checked, so both statements are proved. ■

7A.2 – Counterexamples

(i) $n = 1$: $1^2 = 1$. (Also $n = 0$.)
(ii) $m = 2, n = -2$: $m^2 = n^2 = 4$ but $m \neq n$.
(iii) $x = \frac{1}{2}$: $x^3 = \frac{1}{8} < \frac{1}{4} = x^2$.
(iv) $f(x) = e^x$: $f'(x) = e^x = f(x)$.
(v) Skew lines: the line $\{(t, 0, 0)\}$ (the x-axis) and the line $\{(0, 1, t)\}$ never meet (the first has $y = 0$, the second $y = 1$) and are not parallel (directions $(1, 0, 0)$ and $(0, 0, 1)$).

7A.3 – $3 \mid n^3 - n$

Every integer has the form $3k$, $3k+1$ or $3k+2$, and $n^3 - n = n(n-1)(n+1)$.
Case $n = 3k$: the factor n is divisible by 3. *Case* $n = 3k+1$: the factor $n - 1 = 3k$ is divisible by 3.
Case $n = 3k+2$: the factor $n + 1 = 3(k+1)$ is divisible by 3.
In every case 3 divides the product. ■

7A.4 – max and min

Case 1: $x \ge y$. Then $\max(x, y) = x$ and $\min(x, y) = y$, so the sum is $x + y$.
Case 2: $x < y$. Then $\max(x, y) = y$ and $\min(x, y) = x$, so the sum is $y + x = x + y$. ■

7A.5 – A constructive existence proof

$n = 1$ works: the only positive integer not exceeding 1 is 1 itself. ($n = 1$ is the only one: $1+2+\cdots+n = \frac{n(n+1)}{2} = n$ gives $n = 1$.) ■

8 Common Mistakes in Proofs

Each of the following "proofs" looks correct at first sight but contains an error. Learn to spot these — examiners often ask "find the error".

Example 8.1 – Dividing by zero: "$1 = 2$"

Find the error in this "proof" that $1 = 2$.

Step 1. Let $a = b$, where a and b are positive integers.
Step 2. Multiply both sides by a: $a^2 = ab$.
Step 3. Subtract b^2: $a^2 - b^2 = ab - b^2$.
Step 4. Factorise: $(a - b)(a + b) = b(a - b)$.
Step 5. Divide both sides by $a - b$: $a + b = b$.
Step 6. Replace a by b: $2b = b$, so $2 = 1$.

Solution. Step 5 is wrong. Since $a = b$, the number $a - b$ is **zero**, and we cannot divide by zero. Every other step is correct.

Example 8.2 – Using the converse (affirming the conclusion)

Find the error: "*Theorem.* If n^2 is positive, then n is positive. *Proof.* Suppose n^2 is positive. Because the conditional 'if n is positive, then n^2 is positive' is true, we conclude that n is positive."

Solution. The proof uses $p \to q$ ("$n > 0 \Rightarrow n^2 > 0$") together with q ("$n^2 > 0$") to conclude p. That is the fallacy of **affirming the conclusion** (affirming the consequent). In fact the "theorem" is false: $n = -1$ gives $n^2 = 1 > 0$ but $n < 0$.

Example 8.3 – Circular reasoning

Find the error: "*Theorem.* If n^2 is even, then n is even. *Proof.* Suppose n^2 is even. Then $n^2 = 2k$ for some integer k. Let $n = 2l$ for some integer l. This shows that n is even."

Solution. The step "let $n = 2l$" *assumes* that n is even, which is exactly what we are trying to prove. Nothing in the earlier steps gives this. This is **circular reasoning** (begging the question). A correct proof is by contraposition (Example 5.2).

Example 8.4 – Proof by examples

Find the error: "*Theorem.* $n^2 - n + 41$ is prime for every positive integer n. *Proof.* For $n = 1, 2, 3, 4, 5$ the values are 41, 43, 47, 53, 61, all prime. Hence the result holds."

Solution. Checking some cases does not prove a "for all" statement. In fact it is false: for $n = 41$, $41^2 - 41 + 41 = 41^2 = 1681$, which is not prime.

Example 8.5 – Wrong negation

Find the error in the start of this proof by contradiction: "To prove that *every* integer n with n^2 odd is odd, assume that *every* integer n with n^2 odd is even."

Solution. The negation of "for all n, $p(n)$" is "for *some* n, $\neg p(n)$", not "for all n, $\neg p(n)$". The correct start is: "Assume that for *some* integer n, n^2 is odd and n is even."

Checklist before you finish a proof

1. Did I use the *definition* (e.g. $n = 2k + 1$) and not just examples?
2. Is every variable that I introduced (like k) said to be an integer?
3. Did I divide only by numbers that cannot be zero?
4. Did I use only $p \to q$ with p (modus ponens), never $p \to q$ with q?
5. For "if and only if", did I prove both directions?
6. Does my last line state exactly what was to be proved?

9 Solved University Examination Questions

The questions below were set in university examinations or appear in university question banks; the source is shown under each question. Write answers in the same step-by-step way.

Problem U1: $\sqrt{2}$ is irrational — 10 marks

Q. Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.
Source: Anna University, B.E./B.Tech. (CSE) III Sem., Nov/Dec 2021 (MA8351), Q11(a)(ii); IGNOU MCS-013, Dec 2023; Narsimha Reddy Engineering College (JNTUH) question bank.

Solution.
Lemma. *If a^2 is even, then a is even.* Proof by contraposition: if a is odd, $a = 2k+1$, then $a^2 = 2(2k^2+2k)+1$ is odd.
Assumption. Suppose, to the contrary, that $\sqrt{2}$ is rational. Then $\sqrt{2} = \frac{a}{b}$ with integers a, b, $b \neq 0$, and $\frac{a}{b}$ in lowest terms (a, b have no common factor other than 1).
Step 1. Squaring: $2 = \frac{a^2}{b^2}$, so $a^2 = 2b^2$. Hence a^2 is even, and by the lemma a is even: $a = 2c$.
Step 2. Then $4c^2 = 2b^2$, so $b^2 = 2c^2$. Hence b^2 is even, and by the lemma b is even.
Step 3. So 2 divides both a and b. This contradicts the assumption that $\frac{a}{b}$ is in lowest terms.
Conclusion. The assumption is false; hence $\sqrt{2}$ is irrational. ■

Problem U2: $n^2 = m^2$ iff $m = \pm n$ — Long answer

Q. Let m and n be integers. Prove that $n^2 = m^2$ if and only if $m = n$ or $m = -n$.
Source: Anna University, B.E./B.Tech. (CSE) III Sem., Nov/Dec 2021 (MA8351).

Solution. The statement has two parts.
"If" part. Suppose $m = n$ or $m = -n$. If $m = n$, then $m^2 = n^2$. If $m = -n$, then $m^2 = (-n)^2 = n^2$. In both cases $n^2 = m^2$.
"Only if" part. Suppose $n^2 = m^2$. Then $n^2 - m^2 = 0$, i.e. $(n-m)(n+m) = 0$. A product of two integers is 0 only if one factor is 0, so $n - m = 0$ or $n + m = 0$, i.e. $m = n$ or $m = -n$.
Conclusion. Both parts hold, so $n^2 = m^2 \iff (m = n \text{ or } m = -n)$. ■

Problem U3: If $3n+2$ is odd then n is odd — Medium answer

Q. Prove that if n is an integer and $3n+2$ is odd, then n is odd.
Source: Narsimha Reddy Engineering College (JNTUH) question bank, Discrete Mathematics.

Solution.
Method 1 (contraposition). The contrapositive is: *if n is even, then $3n+2$ is even.* Let $n = 2k$. Then $3n+2 = 6k+2 = 2(3k+1)$, which is even. The contrapositive is true, so the statement is true.
Method 2 (contradiction). Assume $3n+2$ is odd and n is even, $n = 2k$. Then $3n+2 = 2(3k+1)$ is even. So $3n+2$ is both odd and even — a contradiction. Hence n is odd. ■

Problem U4: If n^3+5 is odd then n is even — Medium answer

Q. Show that if n is an integer and n^3+5 is odd, then n is even. (Solved here in two ways.)
Source: Kongunadu College of Engineering and Technology (Autonomous), question bank 24MA301, 2025–26.

Solution. First a calculation: if $n = 2k+1$, then

$$n^3 + 5 = (8k^3 + 12k^2 + 6k + 1) + 5 = 8k^3 + 12k^2 + 6k + 6 = 2(4k^3 + 6k^2 + 3k + 3),$$

which is even.
(a) Contraposition. The contrapositive is: *if n is odd, then n^3+5 is even.* This is exactly the calculation above. So the statement is true.

(b) Contradiction. Assume n^3+5 is odd and n is not even, i.e. n is odd. By the calculation, n^3+5 is even. So n^3+5 is both odd and even — a contradiction. Hence n is even. ■

Problem U5: Sum of two rational numbers — Medium answer

Q. Prove that the sum of two rational numbers is rational.

Source: Kongunadu College of Engineering and Technology (Autonomous), question bank 24MA301, 2025–26.

Solution. (Forward proof.) Let r and s be rational. Then $r=\frac{a}{b}$ and $s=\frac{c}{d}$ for integers a,b,c,d with $b\neq 0, d\neq 0$. Hence

$$r+s=\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}.$$

Here $ad+bc$ and bd are integers (sums and products of integers are integers), and $bd\neq 0$. So $r+s$ is a quotient of two integers with non-zero denominator, i.e. $r+s$ is rational. ■

Problem U6: $\sqrt{3}$ is irrational — Long answer

Q. Show that $\sqrt{3}$ is irrational by giving a proof using contradiction.

Source: Kongunadu College of Engineering and Technology (Autonomous), question bank 24MA301, 2025–26.

Solution.

Lemma. *If $3 \mid a^2$, then $3 \mid a$.* By contraposition: if $3 \nmid a$, then $a=3k+1$ or $a=3k+2$, and $a^2=3(3k^2+2k)+1$ or $a^2=3(3k^2+4k+1)+1$; in both cases $3\nmid a^2$.

Assumption. Suppose $\sqrt{3}=\frac{a}{b}$ in lowest terms, with integers a,b, $b\neq 0$.

Step 1. Then $a^2=3b^2$, so $3\mid a^2$ and, by the lemma, $3\mid a$: $a=3c$.

Step 2. Then $9c^2=3b^2$, so $b^2=3c^2$; hence $3\mid b^2$ and $3\mid b$.

Step 3. So 3 is a common factor of a and b, contradicting "lowest terms".

Conclusion. Hence $\sqrt{3}$ is irrational. ■

Problem U7: Direct proof — odd squares — Short answer

Q. Give a direct proof of the theorem: if m is an odd integer, then m^2 is odd.

Source: Kongunadu College of Engineering and Technology (Autonomous), question bank 24MA301, 2025–26.

Solution. Assume m is odd. Then $m=2k+1$ for some integer k, so

$$m^2=4k^2+4k+1=2(2k^2+2k)+1,$$

which is odd because $2k^2+2k$ is an integer. Hence if m is odd, m^2 is odd. ■

Problem U8: Indirect proof — Medium answer

Q. Give an indirect proof of the theorem: if n is an integer and n^2+3 is odd, then n is even.

Source: Kongunadu College of Engineering and Technology (Autonomous), question bank 24MA301, 2025–26.

Solution. An indirect proof proves the contrapositive: *if n is odd, then n^2+3 is even.* Let $n=2k+1$. Then

$$n^2+3=4k^2+4k+1+3=4k^2+4k+4=2(2k^2+2k+2),$$

which is even. The contrapositive is true, so the theorem is true. ■

Problem U9: Explain proof by contradiction — 2 marks

Q. Explain proof by contradiction, with the help of an example.

Source: IGNOU MCS-013, December 2016.

Solution. To prove a statement P by contradiction, we assume that P is false and show that this leads to something impossible (a contradiction). Then P must be true.
Example: "There is no largest integer." Assume there is a largest integer N. Then $N+1$ is an integer and $N+1 > N$, contradicting the choice of N. Hence there is no largest integer. ■

Problem U10: Converse and contrapositive — Short answer

Q. Write the contrapositive and the converse of: "If $2+2=5$, then I am Prime Minister of India."
Source: IGNOU MCS-013, December 2016.

Solution. Let p: $2+2=5$, q: I am Prime Minister of India. The statement is $p \to q$.
Contrapositive ($\neg q \to \neg p$): "If I am not Prime Minister of India, then $2+2 \neq 5$."
Converse ($q \to p$): "If I am Prime Minister of India, then $2+2=5$."
Note: the given statement is true because its hypothesis $2+2=5$ is false; its contrapositive is true too, because a statement and its contrapositive always have the same truth value.

Problem U11: Short note on contrapositive — Short note

Q. Write a short note on the contrapositive.
Source: IGNOU MCS-013, December 2023.

Solution. The **contrapositive** of $p \to q$ is $\neg q \to \neg p$. For example, the contrapositive of "If it rains, the ground is wet" is "If the ground is not wet, it did not rain".
A conditional and its contrapositive are **logically equivalent**: both are false only when p is true and q is false (see the truth table in Section 5). The converse $q \to p$ and the inverse $\neg p \to \neg q$ are *not* equivalent to $p \to q$.
Use in proofs: to prove $p \to q$ we may prove $\neg q \to \neg p$ instead (proof by contraposition). For example, "if n^2 is odd, then n is odd" is proved by showing "if n is even, then n^2 is even".

Problem U12: Contraposition with a product — 5 marks

Q. Using proof by contrapositive, prove that for all integers x and y, if xy is odd, then both x and y are odd.
Source: GGSIPU, B.Tech (CIC) III Sem., End Term Examination, Dec 2024 (CIC-205), Q3(a).

Solution. Let p: xy is odd, q: x and y are both odd. We prove the contrapositive $\neg q \to \neg p$.
Contrapositive. If x is even *or* y is even, then xy is even. (Note: $\neg(x \text{ odd} \wedge y \text{ odd}) \equiv x \text{ even} \vee y \text{ even}$, by De Morgan's law.)
Case 1: x even. Then $x = 2k$, so $xy = 2(ky)$, which is even.
Case 2: y even. Then $y = 2l$, so $xy = 2(xl)$, which is even.
Conclusion. In both cases xy is even, i.e. not odd. The contrapositive is true, so the original statement is true. ■

10 Exercises

Try these yourself first; complete solutions are in Section 11.

10.1 Part A — Short answer (2 marks each)

A1. Define lemma and corollary, with one example each.

A2. What is a conjecture? Name one famous conjecture.

A3. What is an axiom? Give an example.

A4. Write the converse and the contrapositive of "If a number is divisible by 4, then it is even." Which of the three statements are true?

A5. State the difference between proof by contraposition and proof by contradiction.

A6. What is a vacuous proof? Give an example.

A7. Disprove by a counterexample: "Every prime number is odd."

A8. Write "p is necessary for q" and "p is sufficient for q" in symbols.

A9. Distinguish between a constructive and a non-constructive proof of existence.

A10. What is proof by exhaustion? When can it be used?

10.2 Part B — Medium answer (5 / 7 marks)

B1. Prove that the sum of two prime numbers, each larger than 2, is not a prime number.

B2. Prove by contradiction: for all positive real numbers x and y, if the product xy exceeds 25, then $x > 5$ or $y > 5$.

B3. Prove that the product of a non-zero rational number and an irrational number is irrational.

B4. Prove that if a and b are integers and $a^2 + b^2$ is even, then $a + b$ is even.

B5. Prove that for integers m and n, mn is odd if and only if m and n are both odd.

B6. Prove or disprove: "The product of two irrational numbers is irrational."

B7. Find the error: "*Theorem.* If x is real and $x^2 = 4$, then $x = 2$. *Proof.* $x^2 = 4$, so $x = \sqrt{4} = 2$."

B8. Prove that no integer of the form $4k + 3$ is a perfect square.

10.3 Part C — Long answer (10 marks)

C1. (a) Prove by contradiction that $\sqrt{p}$ is irrational for every prime p. (b) Hence prove that $\sqrt{2} + \sqrt{3}$ is irrational.

C2. (a) Show that the equation $x^3 + x + 1 = 0$ has no rational root. (b) Prove that if $a_1, a_2, \ldots, a_n$ are real numbers, at least one of them is greater than or equal to their average.

C3. (a) Show that the following statements about an integer n are equivalent: (i) n is even; (ii) $3n + 2$ is even; (iii) n^2 is even. (b) Prove that for every non-zero real number a there is exactly one real number b with $ab = 1$.

C4. Prove or disprove each statement: (a) $n^2 - n + 41$ is prime for every positive integer n. (b) $2^n - 1$ is prime whenever n is prime. (c) Every odd integer greater than 1 is prime. (d) For every real number x, if $x^2 > 1$ then $x > 1$.

11 Solutions to Exercises

11.1 Part A

A1. A **lemma** is a small result proved to help prove a bigger theorem, e.g. "if n^2 is even then n is even" (used to prove $\sqrt{2}$ irrational). A **corollary** follows easily from a theorem, e.g. "each angle of an equilateral triangle is $60°$" follows from "the angles of a triangle add up to $180°$".

A2. A statement believed to be true but not yet proved. Example: Goldbach's conjecture — every even integer greater than 2 is the sum of two primes.

A3. A statement accepted as true without proof, e.g. $a + b = b + a$ for all real numbers a, b.

A4. Converse: "If a number is even, then it is divisible by 4" — false (2 is even but not divisible by 4). Contrapositive: "If a number is not even, then it is not divisible by 4" — true. The original statement is true.

A5. To prove $p \to q$: by *contraposition* we assume $\neg q$ and must reach $\neg p$; by *contradiction* we assume p and $\neg q$ and may reach *any* contradiction.

A6. A proof of $p \to q$ in which p is false, so the conditional is automatically true. Example: "if $0 > 1$ then $0^2 > 0$" is true because $0 > 1$ is false.

A7. 2 is prime but not odd.

A8. "p is necessary for q": $q \to p$. "p is sufficient for q": $p \to q$.

A9. A *constructive* proof exhibits an actual element (e.g. $1729 = 10^3 + 9^3 = 12^3 + 1^3$); a *non-constructive* proof shows that one must exist without finding it (e.g. the $\sqrt{2}^{\sqrt{2}}$ argument in Section 7).

A10. Proving $\forall x \in S,\ p(x)$ by checking every element of S. It can be used only when S is finite and small.

11.2 Part B

B1. (Forward proof.) Every prime larger than 2 is odd (an even number larger than 2 has the divisor 2). If $p, q > 2$ are primes, then $p + q$ is a sum of two odd integers, hence even, and $p + q > 4$. An even integer greater than 2 has the divisor 2 with $1 < 2 < p + q$, so it is not prime. ■

B2. Assume $xy > 25$ but $x \leq 5$ and $y \leq 5$. Since $x, y > 0$, multiplying gives $xy \leq 5 \cdot 5 = 25$, contradicting $xy > 25$. Hence $x > 5$ or $y > 5$. ■

B3. Let $r \neq 0$ be rational and x irrational. Assume, to the contrary, that $rx = s$ is rational. Then $x = \frac{s}{r}$ (allowed, as $r \neq 0$) is a quotient of two rationals, hence rational — a contradiction. So rx is irrational. ■

B4. (Contraposition.) Suppose $a + b$ is odd. Then one of a, b is even and the other odd; say $a = 2k, b = 2l + 1$. Then $a^2 + b^2 = 4k^2 + 4l^2 + 4l + 1 = 2(2k^2 + 2l^2 + 2l) + 1$ is odd. So "$a + b$ odd $\Rightarrow a^2 + b^2$ odd", which proves the statement. ■

B5. ($\Leftarrow$) If $m = 2k + 1$ and $n = 2l + 1$, then $mn = 2(2kl + k + l) + 1$ is odd.
($\Rightarrow$, contraposition) If m or n is even, say $m = 2k$, then $mn = 2(kn)$ is even.
Hence mn is odd iff m and n are both odd. ■

B6. False. $\sqrt{2}$ is irrational, but $\sqrt{2} \cdot \sqrt{2} = 2$ is rational. This counterexample disproves the statement.

B7. From $x^2 = 4$ we get $x = 2$ *or* $x = -2$; the step "$x = \sqrt{4}$" ignores the negative root (in fact $\sqrt{x^2} = |x|$). The "theorem" is false: $x = -2$ is a counterexample.

B8. By Section 7, the square of every integer has the form $4k$ or $4k + 1$, i.e. leaves remainder 0 or 1 on division by 4. A number of the form $4k + 3$ leaves remainder 3, so it cannot be a perfect square. ■ (This is a corollary.)

11.3 Part C

C1. (a) We use the fact: *if a prime p divides a product ab, then $p \mid a$ or $p \mid b$.* Assume $\sqrt{p} = \frac{a}{b}$ in lowest terms. Then $a^2 = pb^2$, so $p \mid a^2$ and hence $p \mid a$: $a = pc$. Then $p^2c^2 = pb^2$, so $b^2 = pc^2$; hence $p \mid b^2$ and $p \mid b$. So p is a common factor of a and b — a contradiction. Hence $\sqrt{p}$ is irrational. ■

(b) Assume $\sqrt{2} + \sqrt{3} = r$ is rational; note $r > 0$. Then $\sqrt{3} = r - \sqrt{2}$. Squaring: $3 = r^2 - 2r\sqrt{2} + 2$, so

$$\sqrt{2} = \frac{r^2 - 1}{2r},$$

which is rational (as $r \neq 0$). This contradicts part (a) with $p = 2$. Hence $\sqrt{2} + \sqrt{3}$ is irrational. ■

C2. (a) Assume $x = \frac{a}{b}$ is a rational root in lowest terms (a, b not both even). Multiplying $\frac{a^3}{b^3} + \frac{a}{b} + 1 = 0$ by b^3 gives

$$a^3 + ab^2 + b^3 = 0.$$

Case 1: a, b both odd. Then a^3, ab^2, b^3 are all odd, and odd + odd + odd is odd, so the sum is not 0.
Case 2: a even, b odd. Then a^3 and ab^2 are even and b^3 is odd, so the sum is odd.
Case 3: a odd, b even. Then a^3 is odd and ab^2, b^3 are even, so the sum is odd.
In every case the sum is odd, so it cannot equal 0 — a contradiction. Hence there is no rational root. ■

(b) Let $A = \frac{a_1 + a_2 + \cdots + a_n}{n}$. Assume, to the contrary, that $a_i < A$ for every i. Adding these n inequalities gives $a_1 + \cdots + a_n < nA = a_1 + \cdots + a_n$ — a contradiction. Hence some $a_i \geq A$. ■

C3. (a) We prove (i)→(ii)→(iii)→(i).
(i)→(ii): $n = 2k \Rightarrow 3n + 2 = 6k + 2 = 2(3k + 1)$, even.
(ii)→(iii): if $3n + 2$ is even, then $3n = (3n + 2) - 2$ is even. So n is even (if n were odd, $3n$ would be odd × odd = odd). Then $n^2 = n \cdot n$ is even.
(iii)→(i): if n^2 is even then n is even (Example 5.2).
The cycle is complete, so (i), (ii), (iii) are equivalent. ■

(b) *Existence:* $b = \frac{1}{a}$ gives $ab = 1$. *Uniqueness:* if $ab = 1$ and $ac = 1$, then $b = b \cdot 1 = b(ac) = (ba)c = 1 \cdot c = c$. So there is exactly one such b. ■

C4. All four statements are **false**:
(a) $n = 41$: $41^2 - 41 + 41 = 41^2 = 1681$, which is not prime.
(b) $n = 11$ is prime, but $2^{11} - 1 = 2047 = 23 \times 89$ is not prime.
(c) 9 is odd and greater than 1, but $9 = 3 \times 3$ is not prime.
(d) $x = -2$: $x^2 = 4 > 1$, but $x \not> 1$.

12 One-Page Revision Sheet

Terminology
Axiom: accepted without proof.
Theorem: proved important statement.
Lemma: helper result. Corollary: easy consequence.
Conjecture: believed true, not yet proved.
Counterexample: one case that makes a $\forall$ false.

Definitions to use
Even: $n = 2k$. Odd: $n = 2k + 1$.
$a \mid b$: $b = ak$. Rational: $\frac{a}{b}$, $b \neq 0$, lowest terms.
odd$\times$odd $=$ odd; odd$+$odd $=$ even.

Forward (direct) proof of $p \to q$
Assume $p \Rightarrow$ steps $\Rightarrow q$.
Find steps backwards if needed; write forwards.

Proof by contradiction
Assume the statement is false ($p \wedge \neg q$);
reach something impossible ($r \wedge \neg r$).
Use for: irrational, infinitely many, "no ...".
Classic: $\sqrt{2}$, $\sqrt{3}$, infinitely many primes.

Proof by contraposition
$p \to q \equiv \neg q \to \neg p$: assume $\neg q$, show $\neg p$.
Use when $\neg q$ gives an equation (e.g. $n = 2k$).
Contrapositive of $p \to (q \vee r)$: $(\neg q \wedge \neg r) \to \neg p$.

Necessity and sufficiency
p sufficient for $q = q$ necessary for $p = p \to q$.
$p \leftrightarrow q$: prove $p \to q$ *and* $q \to p$.
"if" part: $q \to p$; "only if" part: $p \to q$.
Many statements: prove a cycle $p_1 \to p_2 \to \ldots \to p_1$.
A chain of $\iff$ needs every step reversible.

Other methods
Cases: $(p_1 \vee p_2) \to q \equiv (p_1 \to q) \wedge (p_2 \to q)$.
Exhaustion: check all elements of a small finite set.
Existence: one example (constructive) or not.
Uniqueness: $p(x) \wedge p(y) \Rightarrow x = y$.
Vacuous: p false. Trivial: q true.

Common mistakes
Dividing by zero; using the converse;
assuming what is to be proved (circular);
proving "for all" by examples;
wrong negation of $\forall$ / $\exists$.

Exam tips
Name the method; write the assumption;
give a reason for every step; end with ■.

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